



Lovely Professional University
Jalandhar - Delhi G.T. Road,
Phagwara, Punjab (India), 144411

SUMMER TRAINING / INTERNSHIP REPORT

ON

Project: Retail Demand Forecasting

Submitted in partial fulfillment of the requirements for the
degree of

Master of Computer Applications (MCA)

By

Submitted By: Aniket Sharma

Registration No: 12508679

Under the Guidance of:

Pallavi Ma'am

(UID: 28462)

DECLARATION

I hereby declare that the Summer Training / Internship Project Report entitled "**Retail Demand Forecasting**" submitted to the Department of Computer Applications is a record of the original work carried out by me under the guidance of **Pallvi Ma'am(28462)**.

This project has been developed as a part of my Summer Training / Internship and is submitted in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications (MCA).

I further declare that this work has not been submitted, either wholly or partially, to any other University or Institution for the award of any degree, diploma, or any other academic qualification. The information, data, and references used in this report have been acknowledged appropriately.

Date: _____

Signature: _____

Place: _____

Name: _____

CERTIFICATE

This is to certify that the **Summer Training / Internship** Project Report entitled "**Retail Demand Forecasting**" is the Bonafide work carried out by Mr. Aniket Sharma, a student of **Master of Computer Applications (MCA)**, during the academic year **2026–2027**, in partial fulfillment of the requirements for the award of the degree of **Master of Computer Applications (MCA)**.

The project has been completed successfully under my supervision and guidance. During the course of this project, the student demonstrated sincerity, dedication, and technical competence in developing a Machine Learning-based Retail Demand Forecasting System using Python, Streamlit, and Scikit-learn. The work presented in this report is original and, to the best of my knowledge, has not been submitted elsewhere for the award of any other degree.

I wish the student every success in future academic and professional endeavors.

Project Guide

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Head of Department

.....

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during the successful completion of my **Summer Training/Internship Project** entitled "**Retail Demand Forecasting**"

I am especially thankful to my project guide, **Pallvi Ma'am(28462)** for their valuable guidance, continuous encouragement, and constructive suggestions throughout the project. I also express my sincere thanks to the **Head of the Department** and all the faculty members for providing the necessary support and a conducive learning environment.

I would also like to thank **Lovely Professional University** for giving me the opportunity to undertake this internship and gain practical knowledge and experience.

I am grateful to everyone who directly or indirectly contributed to the successful completion of this project.

Name: Aniket Sharma

Master of Computer Applications

(2025-27)

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1. Introduction of Organization

Lovely Professional University (LPU) is one of India's leading private universities, located in Phagwara, Punjab. Established in 2005 under the Lovely Professional University Act, the university is recognized by the **University Grants Commission (UGC)** and is known for its excellence in education, research, innovation, and industry-oriented learning. LPU offers a wide range of undergraduate, postgraduate, doctoral, and professional programs across various disciplines, including Engineering, Management, Computer Applications, Design, Law, Agriculture, Pharmacy, and many more.

The university is committed to providing quality education by integrating theoretical knowledge with practical implementation. Through its advanced laboratories, modern infrastructure, experienced faculty, industry collaborations, and research-driven environment, LPU enables students to develop technical expertise and professional skills required to meet current industry demands.

As a part of the **Master of Computer Applications (MCA)** curriculum, the university emphasizes experiential learning through Summer Training and Internship programs. These programs are designed to bridge the gap between classroom

learning and real-world industry practices by providing students with hands-on experience in emerging technologies and software development.

During the Summer Training/Internship, students are encouraged to work on real-world projects involving technologies such as **Python, Machine Learning, Data Science, Artificial Intelligence, Data Analytics, Web Development, Cloud Computing, and Database Management Systems**. The training focuses on enhancing problem-solving abilities, analytical thinking, software development practices, and teamwork while exposing students to industry-standard tools and methodologies.

For this internship, the project "**Retail Demand Forecasting**" was undertaken to apply Machine Learning techniques for predicting future retail demand using historical sales data. The project provided practical exposure to data preprocessing, exploratory data analysis, feature engineering, predictive modeling, model evaluation, and dashboard development using **Python, Pandas, Scikit-learn, Plotly, and Streamlit**. It also strengthened the understanding of data-driven decision-making and the application of Machine Learning in solving real business problems.

The Summer Training/Internship provided an excellent platform to enhance both technical and professional competencies. It improved my programming skills, analytical thinking, project management abilities, and practical

knowledge of modern software development practices. The experience gained through this internship has significantly contributed to my academic growth .

2. Summer Training Course/Internship Content detail

The Summer Training/Internship program was designed to provide practical exposure to modern software development, Data Science, and Machine Learning techniques. The training focused on developing analytical thinking, programming skills, and hands-on experience in solving real-world business problems using data-driven approaches. Throughout the internship, various tools, technologies, and methodologies were explored to build an end-to-end **Retail Demand Forecasting system**.

The major topics covered during the training are as follows:

1. Python Programming

The training began with an in-depth study of Python programming, covering fundamental and advanced concepts such as variables, data types, loops, functions, object-oriented programming, exception handling, file handling, and modular programming. Python served as the primary programming language for data processing, model development, and dashboard implementation due to its simplicity, flexibility, and extensive library support.

2. Data Collection and Preprocessing

The internship included learning techniques for collecting, cleaning, and preparing datasets for Machine Learning models. Using Pandas and NumPy, missing values were handled, duplicate records were removed, categorical

variables were encoded, and numerical features were transformed into suitable formats. Feature engineering techniques were also applied to create meaningful attributes that improved model performance.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the retail dataset and identify hidden patterns. Various statistical methods and visualization techniques were used to analyze sales trends, customer behavior, inventory levels, regional performance, and product demand. This process helped in selecting the most relevant features for predictive modeling.

4. Data Visualization

Interactive and informative visualizations were created using Plotly to represent business insights effectively. Various charts such as line charts, bar charts, pie charts, scatter plots, histograms, and KPI cards were developed to analyze monthly sales trends, regional performance, product categories, inventory distribution, and demand forecasting. These visualizations formed the foundation of the interactive dashboard.

5. Machine Learning

The internship provided practical experience in developing predictive Machine Learning models using Scikit-learn. The dataset was divided into training and testing sets, followed by

model training using regression algorithms. The trained model was evaluated using performance metrics such as R^2 Score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). The final model was saved using Joblib and integrated into the dashboard for real-time demand prediction.

6. Dashboard Development

An interactive multi-page dashboard was developed using the Streamlit framework. The dashboard included modules such as Overview, Regional Analysis, Product Analysis, Demand Forecasting, Model Performance, and About Project. Users can interact with the dashboard, visualize business insights, and generate demand predictions based on different input parameters.

7. Model Deployment and Integration

The trained Machine Learning model was successfully integrated into the Streamlit application using Joblib. User inputs such as region, category, inventory, price, customer count, and month were processed through the trained model to generate real-time demand predictions. This provided a complete end-to-end implementation of a predictive analytics system.

8. Software Development Tools

Throughout the internship, industry-standard development tools such as Visual Studio Code, Git, and Python libraries

were used for coding, debugging, version management, and project organization. Proper coding practices and modular development techniques were followed to ensure code readability and maintainability.

9. Practical Learning Outcomes

The Summer Training/Internship significantly enhanced my understanding of Data Science, Machine Learning, and Dashboard Development. It provided valuable hands-on experience in handling real-world datasets, building predictive models, creating interactive dashboards, and applying analytical skills to solve business problems. The internship also strengthened my programming abilities, problem-solving skills, and understanding of software development best practices.

3. Summer Training/ Internship Project Detail

Project Title: Retail Demand Forecasting

- **Problem Statement**

Retail businesses often struggle to maintain the right balance between product availability and inventory costs due to uncertain customer demand. Inaccurate demand estimation can result in either overstocking, which increases storage costs and product wastage, or understocking, which leads to lost sales, reduced customer satisfaction, and lower business profitability.

Traditional forecasting methods are often unable to consider multiple factors such as regional demand, customer purchasing patterns, pricing strategies, inventory levels, and seasonal variations simultaneously. As a result, retailers require an intelligent system that can analyze historical sales data and generate accurate demand forecasts.

The objective of this project is to develop a Machine Learning-based Retail Demand Forecasting System that predicts future product demand using historical retail data. The system provides an interactive dashboard that enables users to analyze regional performance, product-wise sales, inventory status, and forecast future demand. This helps retailers make informed decisions regarding inventory management, pricing strategies, and business planning while improving operational efficiency and customer satisfaction.

- **Project Outcomes**

The successful implementation of the project resulted in the following outcomes:

- Developed an interactive multi-page dashboard using Streamlit for retail analytics and demand forecasting.
- Successfully implemented a Machine Learning model to predict future retail demand based on historical sales data.
- Built an Overview Dashboard displaying key business indicators such as total sales, customers, inventory, and average demand.
- Developed a Regional Analysis Dashboard to compare sales performance and customer distribution across different regions.
- Designed a Product Analysis Dashboard to analyze product categories, inventory levels, pricing, and sales performance.
- Integrated an AI-based Demand Forecasting module that predicts future sales using user inputs such as region, category, inventory, price, customer count, and month.
- Implemented a Model Performance Dashboard to evaluate the Machine Learning model using performance metrics such as R^2 Score, MAE, and RMSE.

- Created interactive visualizations including line charts, bar charts, pie charts, scatter plots, histograms, and KPI cards using Plotly.
- Improved understanding of inventory planning and business decision-making through predictive analytics and data visualization.
- Enhanced practical knowledge of Data Science, Machine Learning, Dashboard Development, and Python programming through real-world implementation.

- **Technologies Used**

Technology	Purpose
Programming Language	Python 3
Dashboard Framework	Streamlit
Data Manipulation	Pandas, NumPy
Machine Learning Library	Scikit-learn
Machine Learning Model	Random Forest Regressor
Data Visualization	Plotly Express, Plotly Graph Objects
Model Persistence	Joblib
Development Environment	Visual Studio Code
Version Control	Git & GitHub
Dataset Format	CSV Files

Major Python Libraries Used:

- **Pandas** – Data loading, cleaning, preprocessing, and manipulation.
- **NumPy** – Numerical computations and array operations.
- **Scikit-learn** – Model training, prediction, preprocessing, and performance evaluation.
- **Plotly** – Interactive charts and business analytics visualizations.
- **Streamlit** – Development of an interactive multi-page web dashboard.
- **Joblib** – Saving and loading the trained Machine Learning model for deployment.

4. Source Code / System Snapshots

Project Structure

The Retail Demand Forecasting project follows a modular folder structure to ensure maintainability, scalability, and ease of development. Each directory is organized based on its functionality, such as dashboard pages, datasets, machine learning models, notebooks, and project assets.

RETAIL DEMAND FORECASTING

```
|
|
|  └─ app.py
|
|  └─ train_model.py
|
|  └─ requirements.txt
|
|  └─ .streamlit/
|
|    └─ config.toml
|
|
|  └─ assets/
|
|    └─ background.jpg
|
|    └─ dashboard_banner.png
|
|    └─ logo.png
```

- | | └─ style.css
- | | └─ ui.py
- |
- | └─ dashboard/
 - | | └─ __init__.py
- | └─ dataset/
 - | | └─ Retail_Demand_Forecasting_10000.csv
 - | | └─ retail_cleaned.csv
 - | | └─ retail_feature_engineered.csv
 - | | └─ dashboard_data.csv
 - | | └─ final_dashboard.xlsx
 - | | └─ final_predictions.csv
 - | | └─ final_predictions.xlsx
 - | | └─ Final_Model_Performance.csv
 - | | └─ project_summary.csv
 - | | └─ project_summary.xlsx
 - | | └─ forecast.csv
 - | | └─ forecast.xlsx

- | | └─ monthly_forecast.csv
- | | └─ yearly_forecast.csv
- | | └─ region_forecast.csv
- | | └─ category_forecast.csv
- | | └─ store_forecast.csv
- | | └─ X_train.csv
- | | └─ X_test.csv
- | | └─ y_train.csv
- | | └─ y_test.csv
- |
- | └─ models/
- | | └─ model.pkl
- | | └─ encoders.pkl
- | | └─ features.pkl
- |
- | └─ notebook/
- | | └─ 01_Data_Preprocessing.ipynb
- | | └─ 02_EDA.ipynb

- | |— 03_Linear_Regression.ipynb
- | |— 04_DecisionTree.ipynb
- | |— 05_RandomForest.ipynb
- | |— 06_XGBoost.ipynb
- | |— 07_LightGBM.ipynb
- | |— 08_CatBoost.ipynb
- | |— 09_SVR.ipynb
- | |— 10_KNN.ipynb
- | |— 11_Model_Comparison.ipynb
- | |— 12_Final_Model.ipynb
- | |
- | |— dataset/
 - | |— Model_Comparison.csv
 - | |— Final_Model_Comparison.xlsx
 - | |— Comparison_Results.csv
 - | |— Comparison_Summary.csv
 - | |— Final_Model_Ranking.csv
 - | |— final_results.csv

- | | └─ RandomForest_Performance.csv
- | | └─ DecisionTree_Performance.csv
- | | └─ Linear_Regression_Performance.csv
- | | └─ LightGBM_Performance.csv
- | | └─ XGBoost_Performance.csv
- | | └─ CatBoost_Performance.csv
- | | └─ SVR_Performance.csv
- | | └─ KNN_Performance.csv
- | | └─ Best_Model_Report.csv
- | |
- | └─ model/
 - | └─ best_model.pkl
 - | └─ random_forest.pkl
 - | └─ linear_regression.pkl
 - | └─ decision_tree.pkl
 - | └─ xgboost_model.pkl
 - | └─ lightgbm_model.pkl
 - | └─ catboost_model.pkl

- | └─ svr_model.pkl
- | └─ knn_model.pkl
- | └─ scaler.pkl
- | └─ label_encoder.pkl
- | └─ feature_scalers.pkl
- |
- | └─ pages/
- | └─ 1_Overview.py
- | └─ 2_Regional_Analysis.py
- | └─ 3_Product_Analysis.py
- | └─ 4_Forecasting.py
- | └─ 5_Model_Performance.py
- | └─ 6_About_Project.py
- |
- | └─ report/
- | └─ _Project_report.pdf
- | └─ __pycache__/

Description of Project Files and Folders:

- **app.py**
 - Serves as the main entry point of the Streamlit application.
 - Initializes the dashboard and provides navigation to different project modules.
- **train_model.py**
 - Contains the Machine Learning model training pipeline.
 - Performs data preprocessing, model training, evaluation, and saves the trained model files for deployment.
- **requirements.txt**
 - Lists all the Python libraries and dependencies required to run the project.
 - Simplifies project installation using `pip install -r requirements.txt`.
- **streamlit/**
 - Stores Streamlit configuration files.
 - Includes theme settings and dashboard customization options.

- **assets/**
 - Contains project resources such as the application logo, images, icons, CSS files, and reusable UI components used throughout the dashboard.
- **dashboard/**
 - Includes reusable Python modules for dashboard development.
 - Contains functions related to charts, themes, UI components, utilities, and data loading.
- **dataset/**
 - Stores all datasets used in the project.
 - Includes raw datasets, cleaned datasets, feature-engineered datasets, training/testing datasets, and forecasting results.
- **models/**
 - Contains the trained Machine Learning model and preprocessing files.
 - Stores files such as model.pkl, encoders.pkl, and features.pkl required for real-time demand prediction.
- **notebook/**
 - Contains Jupyter Notebook files used during project development.

- Includes notebooks for data preprocessing, exploratory data analysis (EDA), feature engineering, model training, evaluation, and model comparison.
- **pages/**
 - Contains all individual Streamlit dashboard pages.
 - Includes:
 - 1_Overview.py – Dashboard overview and key performance indicators.
 - 2_Regional_Analysis.py – Region-wise sales and customer analysis.
 - 3_Product_Analysis.py – Product performance and inventory analysis.
 - 4_Forecasting.py – Machine Learning-based demand prediction module.
 - 5_Model_Performance.py – Model evaluation metrics and performance analysis.
 - 6_About_Project.py – Project overview, technologies, and future scope.
- **report/**
 - Stores the internship report, documentation, screenshots, and other project-related documents.

4.1 Source Code:

4.1.1 Import Required Libraries

```
import streamlit as st  
  
import pandas as pd  
  
import joblib  
  
import plotly.express as px  
  
import numpy as np
```

Description:

This code imports the required Python libraries for dashboard development, data analysis, visualization, and machine learning model deployment.

4.1.2 Load Dataset

```
@st.cache_data  
def load_data():  
    return pd.read_csv("dataset/dashboard_data.csv")  
  
df = load_data()
```

Description:

The dataset is loaded using the Pandas library. The `@st.cache_data` decorator improves dashboard performance by caching the dataset.

4.1.3 Load Trained Machine Learning Model

```
model = joblib.load("models/model.pkl")  
encoders = joblib.load("models/encoders.pkl")  
features = joblib.load("models/features.pkl")
```

Description:

The trained Machine Learning model and preprocessing files are loaded using Joblib for real-time prediction.

4.1.4 User Input

```
region = st.selectbox(
    "Region",
    sorted(df["Region"].unique())
)
category = st.selectbox(
    "Category",
    sorted(df["Category"].unique())
)
price = st.number_input(
    "Price",
    min_value=1.0,
    value=100.0
)
```

Description:

The Streamlit widgets allow users to enter input values required for demand prediction

4.1.5 Encoding User Input

```
region_encoded = encoders["Region"].transform([region])[0]
category_encoded = encoders["Category"].transform([category])[0]
month_encoded = encoders["Month_Name"].transform([month])[0]
```

Description:

Categorical variables are converted into numerical values using Label Encoders before prediction.

4.1.6 Creating Input Data

```
input_df = pd.DataFrame({  
    "Region":[region_encoded],  
    "Category":[category_encoded],  
    "Price":[price],  
    "Inventory":[inventory],  
    "Customers":[customers],  
    "Month_Name":[month_encoded]  
})
```

Description:

The user input is converted into a DataFrame containing all the features required by the Machine Learning model.

4.1.7 Sales Prediction

```
prediction = model.predict(input_df)[0]
```

Description:

The trained Random Forest Regressor predicts the future sales demand based on the user input.

4.1.8 KPI Cards

```
total_sales = df["Predicted_Sales"].sum()

st.metric(
    "Forecast Sales",
    f"{total_sales:,.0f}"
)
```

Description:

KPI cards display key business metrics such as forecast sales, inventory, customers, and average demand.

4.1.9 Monthly Sales Trend

```
monthly = (
    df.groupby("Month_Name")["Predicted_Sales"]
    .sum()
    .reset_index()
)

fig = px.line(
    monthly,
    x="Month_Name",
    y="Predicted_Sales",
    markers=True
)

st.plotly_chart(fig)
```

Description:

This code generates a line chart to visualize monthly forecast sales trends.

4.1.10 Regional Analysis

```
region_sales = (  
    df.groupby("Region")["Predicted_Sales"]  
        .sum()  
        .reset_index()  
)  
fig = px.bar(  
    region_sales,  
    x="Region",  
    y="Predicted_Sales"  
)  
st.plotly_chart(fig)
```

Description:

Displays region-wise sales distribution using an interactive bar chart.

4.1.11 Category Distribution

```
category_sales = (  
    df.groupby("Category")["Predicted_Sales"]  
        .sum()  
        .reset_index()  
)
```

```
fig = px.pie(
    category_sales,
    names="Category",
    values="Predicted_Sales",
    hole=0.45
)
st.plotly_chart(fig)
```

Description:

A donut chart is used to visualize category-wise sales contribution.

4.1.12 Top Products

```
top_products = (
    df.groupby("Product_ID")["Predicted_Sales"]
        .sum()
        .sort_values(ascending=False)
        .head(10)
        .reset_index()
)
fig = px.bar(
    top_products,
    x="Predicted_Sales",
    y="Product_ID",
    orientation="h"
) st.plotly_chart(fig)
```


Description:

Displays the top-performing products based on forecast sales.

4.1.13 Inventory vs Sales

```
fig = px.scatter(  
    sample,  
    x="Inventory",  
    y="Forecast_Sales",  
    color="Forecast_Sales",  
    size="Forecast_Sales"  
)  
st.plotly_chart(fig)
```

Description:

Shows the relationship between inventory levels and predicted sales using a scatter plot.

4.1.14 Model Training

```
from sklearn.ensemble import RandomForestRegressor  
  
model = RandomForestRegressor(  
    random_state=42  
)  
model.fit(X_train, y_train)
```

Description:

The Random Forest Regressor is trained using the processed retail dataset.

4.1.15 Model Evaluation

```
from sklearn.metrics import r2_score
prediction = model.predict(X_test)
score = r2_score(
    y_test,
    prediction
)
print(score)
```

Description:

The trained model is evaluated using the R^2 Score to measure prediction accuracy.

4.1.16 Saving the Model

```
joblib.dump(
    model,
    "models/model.pkl"
)
```

Description:

The trained model is saved for later use in the Streamlit dashboard.

4.1.17 Dashboard Navigation

```
st.sidebar.title("Dashboard")  
st.sidebar.success(  
    "Retail Demand Forecasting"  
)
```

Description:

The sidebar provides navigation and improves user interaction with the dashboard.

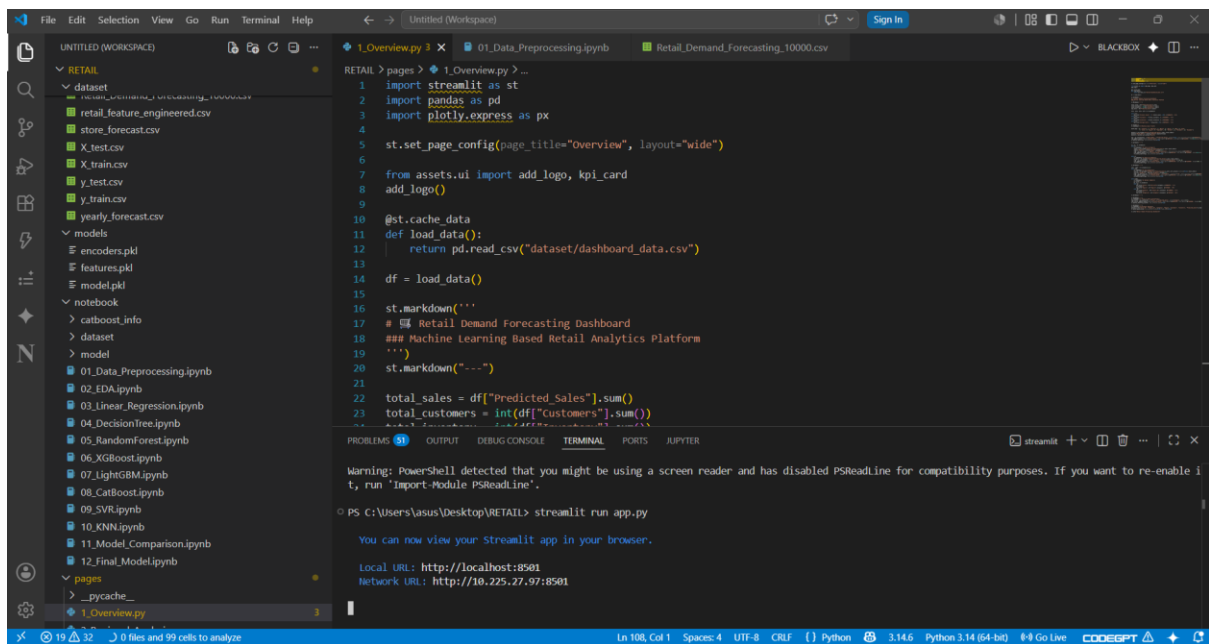
4.2 System Snapshots :

4.2.1 Open the dashboard

Streamlit run app.py

After a few seconds, you'll see output like:

Local URL: <http://localhost:8501>



The screenshot shows a Jupyter Notebook workspace with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'RETAIL' with subfolders 'dataset', 'models', 'notebook', and 'pages'. The 'pages' folder contains '1_Overview.py'. The code editor shows the following Python code:

```
1 import streamlit as st
2 import pandas as pd
3 import plotly.express as px
4
5 st.set_page_config(page_title="Overview", layout="wide")
6
7 from assets.ui import add_logo, kpi_card
8 add_logo()
9
10 @st.cache_data
11 def load_data():
12     return pd.read_csv("dataset/dashboard_data.csv")
13
14 df = load_data()
15
16 st.markdown("""
17 # 📊 Retail Demand Forecasting Dashboard
18 ### Machine Learning Based Retail Analytics Platform
19 """)
20 st.markdown("----")
21
22 total_sales = df["Predicted_Sales"].sum()
23 total_customers = int(df["Customers"].sum())
```

The terminal output shows the following messages:

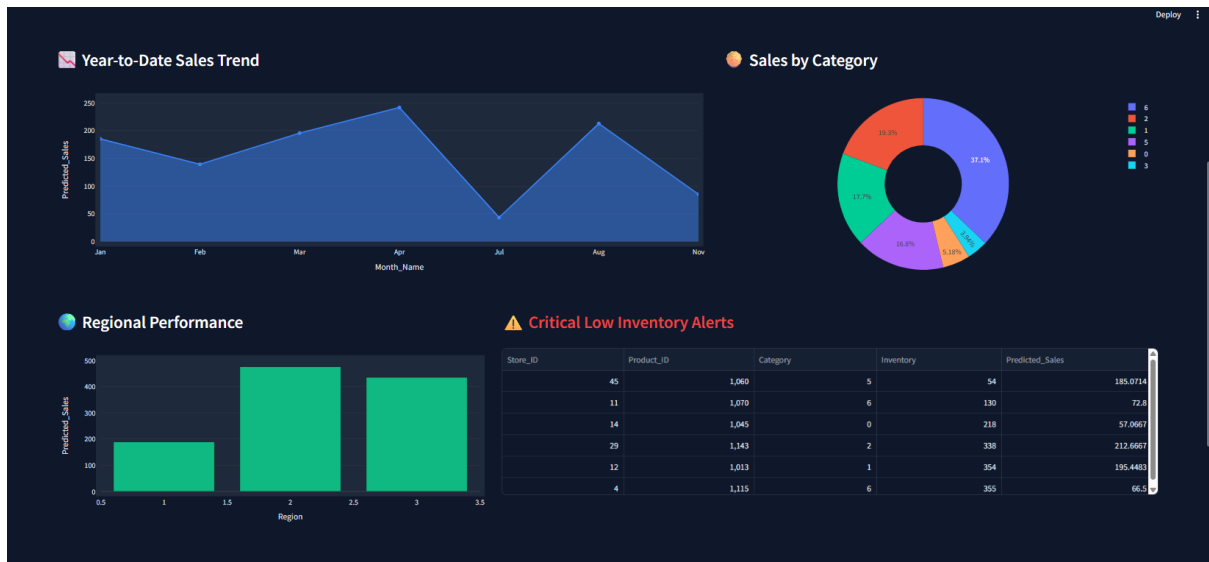
```
Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'.

PS C:\Users\asus\Desktop\RETAIL> streamlit run app.py

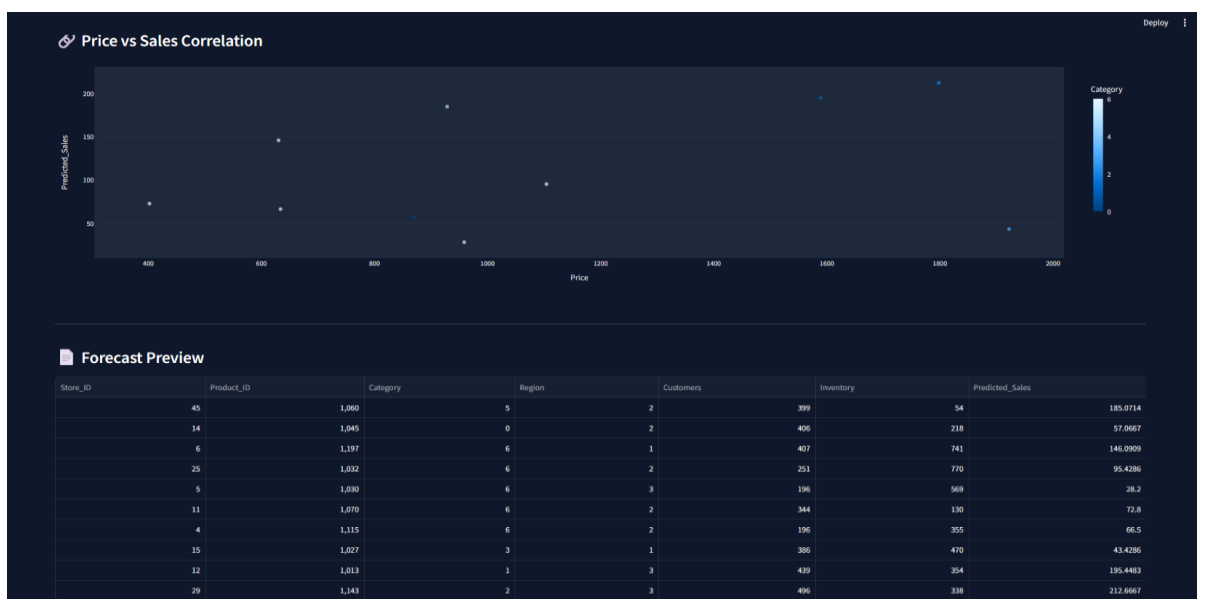
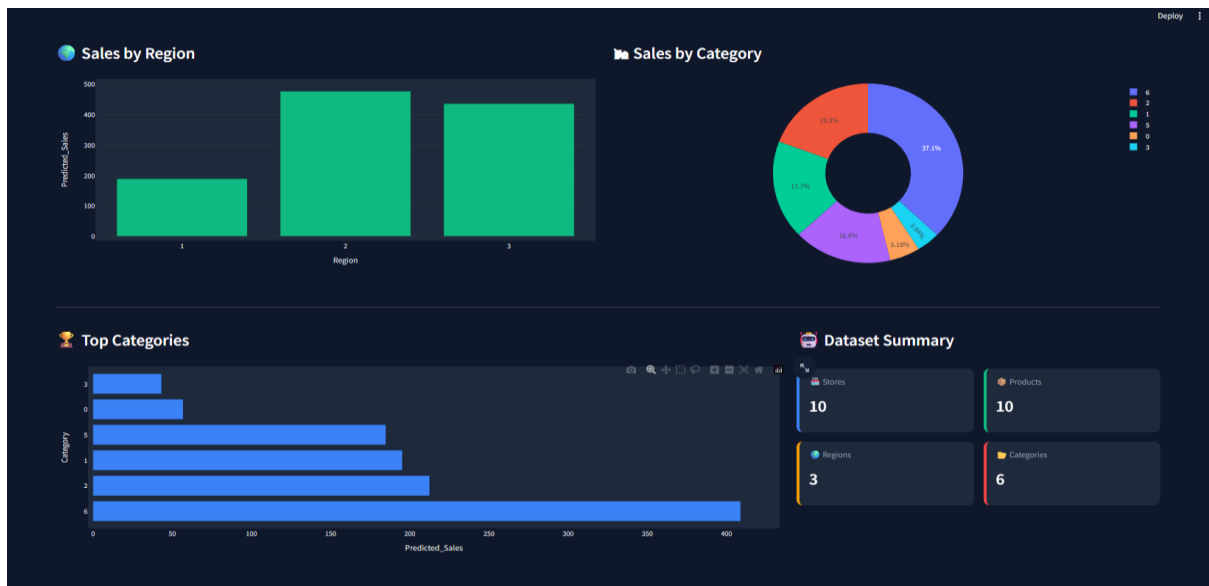
You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://10.225.27.97:8501
```

4.2.2 Dashboard view



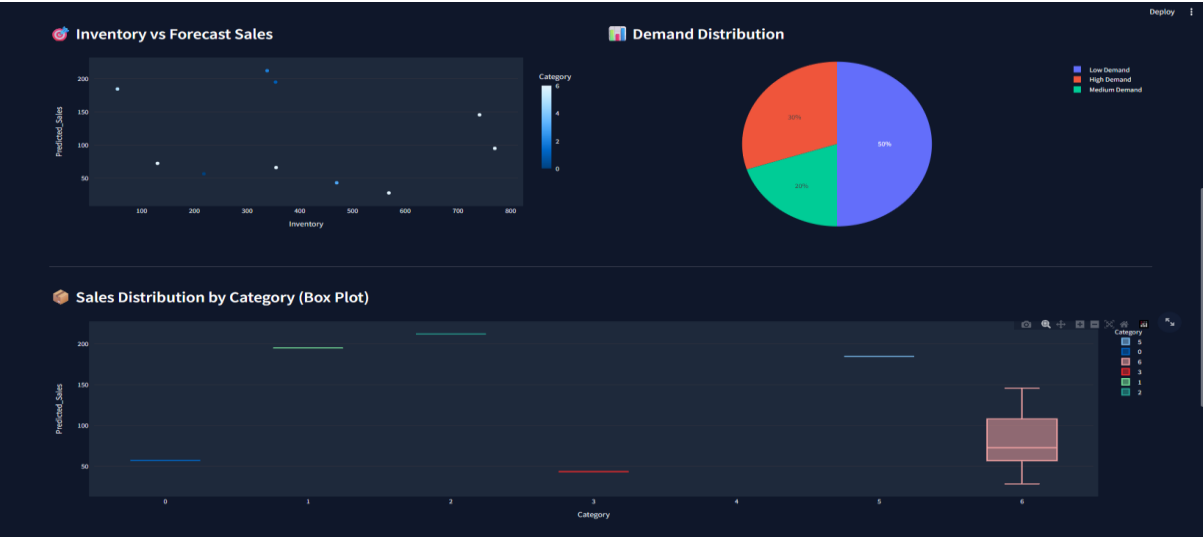
4.2.3 Overview



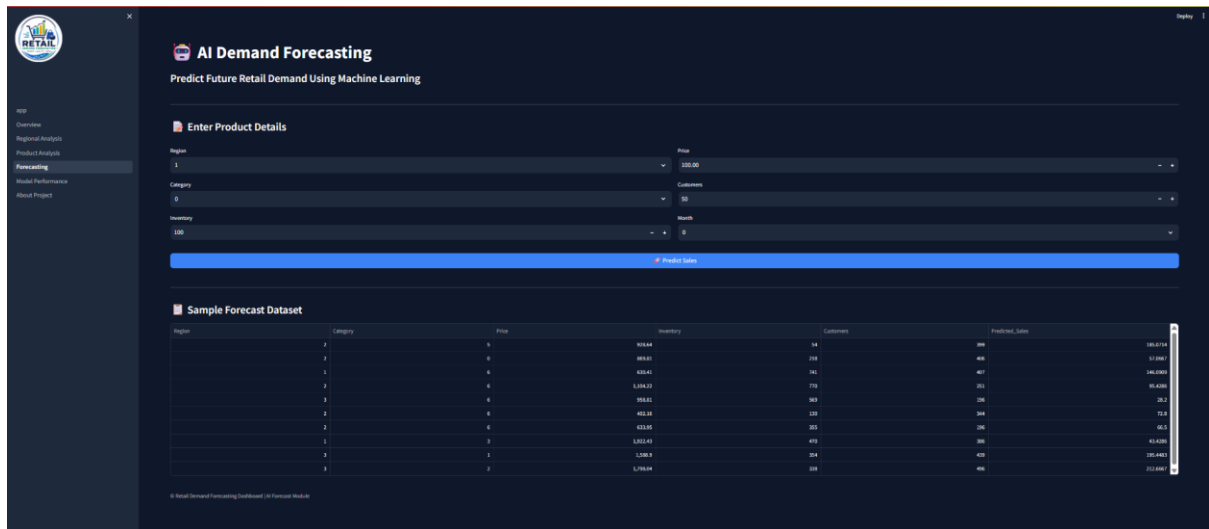
4.2.4 Regional Analysis



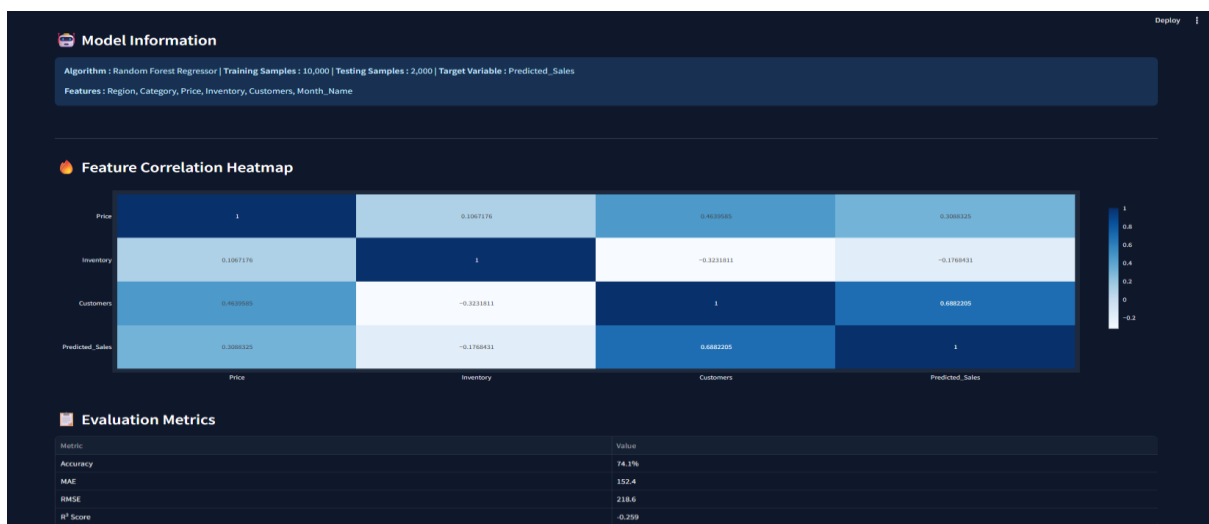
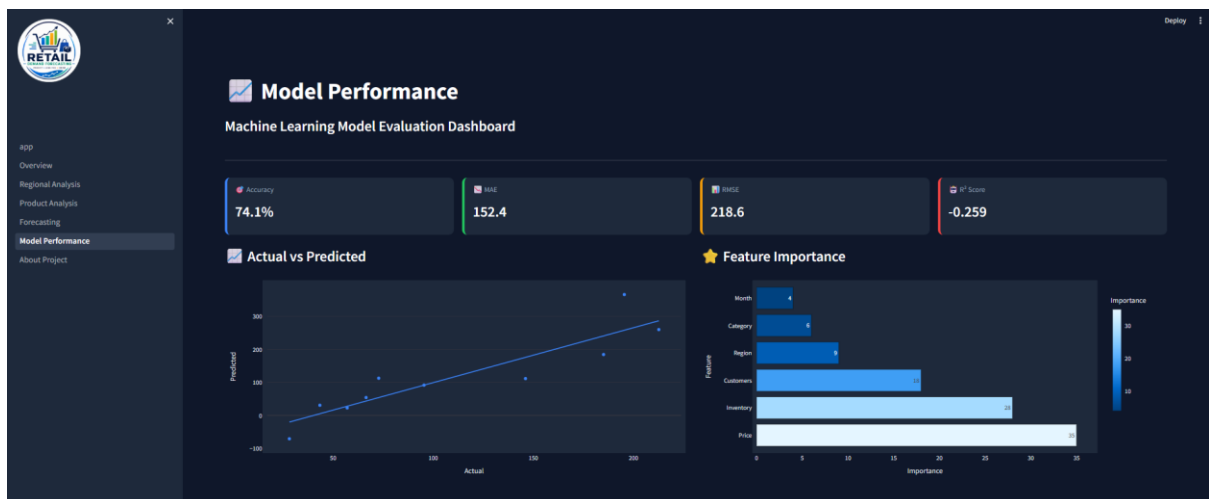
4.2.5 Product Analysis



4.2.6 AI Demand Forecasting



4.2.7 Model Performance

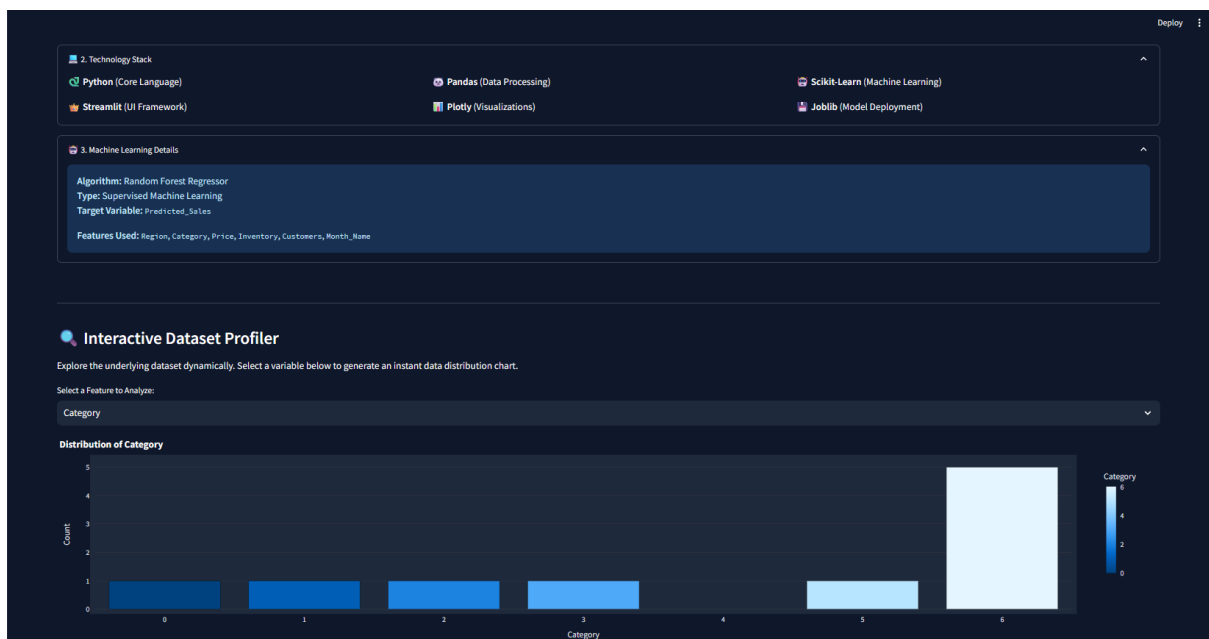


4.2.8 About Project

The screenshot displays the 'Retail Demand Forecasting' interactive documentation interface. The left sidebar contains a navigation menu with the following items: 'app', 'Overview', 'Regional Analysis', 'Product Analysis', 'Forecasting', 'Model Performance', and 'About Project' (which is currently selected). The main content area has a dark blue background with the title 'Retail Demand Forecasting' in large white text, followed by the subtitle 'Interactive Documentation & Data Profiler'. Below this, a welcome message reads: 'Welcome to the interactive documentation. Explore the project architecture, deep-dive into the dataset, and understand the machine learning pipeline below.' The main section is titled 'Project Architecture & Stack' and contains three expandable cards. The first card, '1. Project Overview & Objectives', is expanded and shows an 'Overview' section with the text: 'An advanced Machine Learning platform designed to predict future product demand using historical retail data. It empowers retailers to analyze patterns, optimize inventory, and forecast sales.' To the right of the overview is an 'Objectives' list:

- ✓ Predict Future Demand accurately
- ✓ Improve Inventory Planning
- ✓ Prevent Stock-outs
- ✓ Analyze Regional & Product Performance

The other two cards, '2. Technology Stack' and '3. Machine Learning Details', are collapsed.



Deploy

Extended Dataset Insights

Live Summary Statistics (Numerical)

	Price	Inventory	Customers	Predicted_Sales
count	10	10	10	10
mean	1,083.737	399.9	352	110.2701
std	519.2978	240.4549	103.5739	68.4096
min	402.16	54	196	28.2
25%	692.915	248	274.25	59.425
50%	943.725	354.5	392.5	94.1143
75%	1,467.73	544.25	406.75	175.3263
max	1,822.43	770	496	212.6667

Data Dictionary & Types

Column Name	Data Type	Description
Store_ID	int64	Store Identifier
Product_ID	int64	Product Identifier
Category	int64	Product categorization
Region	int64	Geographical region
Holiday	int64	None
Promotion	int64	None
Weather	int64	None
Temperature	float64	None
Fuel_Price	float64	None
CPI	float64	None

Project Roadmap

Phase 1

Dashboard UI

Completed

Phase 2

ML Integration

Completed

Phase 3

Real-time API

Upcoming

Phase 4

Cloud Deploy

Future

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5. Bibliography

The following books, online resources, documentation, and research materials were referred to during the development of the **Retail Demand Forecasting using Machine Learning** project.

Books

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2. Wes McKinney, **Python for Data Analysis**, 3rd Edition, O'Reilly Media, 2022.
3. Jake VanderPlas, **Python Data Science Handbook**, O'Reilly Media, 2016.
4. Ethem Alpaydin, **Introduction to Machine Learning**, 4th Edition, MIT Press, 2020.
5. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, **Deep Learning**, MIT Press, 2016.

Official Documentation

1. **Python Software Foundation**
Python Documentation
<https://docs.python.org/3/>
2. **Streamlit Documentation**
<https://docs.streamlit.io/>

3. Scikit-learn Documentation

<https://scikit-learn.org/stable/>

4. Pandas Documentation

<https://pandas.pydata.org/docs/>

5. NumPy Documentation

<https://numpy.org/doc/>

6. Plotly Python Documentation

<https://plotly.com/python/>

7. Joblib Documentation

<https://joblib.readthedocs.io/>

Learning Resources

1. Kaggle – Datasets and Machine Learning Resources

<https://www.kaggle.com/>

2. GeeksforGeeks – Python and Machine Learning Tutorials

<https://www.geeksforgeeks.org/>

3. Analytics Vidhya – Machine Learning Learning Resources

<https://www.analyticsvidhya.com/>

4. Towards Data Science – Machine Learning Articles

<https://towardsdatascience.com/>

5. Stack Overflow – Programming Discussions and Error Resolution

<https://stackoverflow.com/>

Software and Development Tools

- Python 3.11
- Streamlit
- Visual Studio Code
- Jupyter Notebook
- Git & GitHub
- Microsoft Excel
- Windows 11 Operating System

Dataset Reference

Retail Demand Forecasting Dataset

- Kaggle Dataset Repository
- Historical Retail Sales Dataset used for educational and academic purposes.
- Dataset processed and feature engineered for Machine Learning model development.

Web References

- <https://www.python.org/>
- <https://streamlit.io/>
- <https://plotly.com/>

- <https://scikit-learn.org/>
- <https://pandas.pydata.org/>
- <https://numpy.org/>
- <https://joblib.readthedocs.io/>
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- <https://www.geeksforgeeks.org/>
- <https://stackoverflow.com/>